BREAKOUT
RADAR

CONFIDENTIAL · INTERNAL LABEL TOOL

A&R INTELLIGENCE / PRODUCT SPECIFICATION

Find the art.
Understand the audience.
Back the future.

An internal intelligence and artist-development system for discovering exceptional talent, tracking fan-created UGC, understanding audience change and making better A&R decisions.

01

UGC Intelligence

Who is using the music, how the content is evolving, and whether it brings new fans.

02

Audience Morphing

What changed in the audience mix, which cohorts are growing and where the opportunity is forming.

03

Artistic Potential

Human-led creative evaluation, catalogue repeatability and identity - in the MVP.

04

Integrity + Backtesting

Filter suspicious growth from the clean shortlist and test recommendations against history.

VERSION 2.0
9 October 2026

Audience evidence. Artistic judgement.
Better decisions, earlier.

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BREAKOUT RADAR

Internal A&R Intelligence, Audience Understanding & Artist Development System

Version 2.0 | 9 October 2026 | Confidential - Internal label strategy

> **Purpose: identify artists with genuine artistic potential before the wider market recognizes them, understand the audiences forming around them, separate authentic growth from suspicious inflation, and turn the evidence into better A&R, signing and artist-development decisions.**

This is an internal tool for a record-label A&R team. It is not a public SaaS product. The label captures its value by discovering, signing and developing exceptional talent, then building sustainable artist brands and businesses.

Executive decision

Breakout Radar should be built as an **internal artist-intelligence and decision-support system**, not as a general analytics dashboard or a product sold to outside customers.

The strongest version combines six disciplines in one workflow:

1. **Talent discovery:** find artists whose current visibility understates their potential.
2. **UGC intelligence:** track how fans and creators actually use an artist's music, including the content types, creators, markets and trends driving that use.
3. **Audience morphing:** measure who an artist is reaching now, which audience cohorts are growing or fading, and how the audience composition differs from the prior snapshot.
4. **Growth integrity:** distinguish likely genuine growth, legitimate paid growth, mixed/uncertain growth, suspicious inflation and provider-confirmed artificial activity.
5. **Artistic potential:** evaluate the craft, originality, identity, catalogue and performance potential required for a lasting career. This is part of Phase 1, not a later add-on.
6. **Prediction and backtesting:** test whether historic signals would have identified promising artists early, then improve models against real outcomes without leaking future information into the past.

The product should not answer only, *"Who has the largest numbers?"* It should answer:

> **"Which artist has the strongest combination of artistic merit, authentic audience formation and future commercial potential relative to their current level of recognition - and what should our label do next?"**

Decisions incorporated from management feedback

- UGC made a first-class intelligence module, distinct from an artist's own social posts.
- Audience composition change becomes a core scorecard and historical dataset.
- Artificial-growth detection becomes a gate on the organic breakout shortlist, with a separate review queue and safeguards against false accusations.

- Artistic-integrity and creative-potential assessment are included in the MVP.
- Backtesting begins during the first build cycle. A first harness is targeted within **10 working days** of data access; an initial baseline report is targeted by **weeks 3-4**, subject to usable historical data.
- Audience understanding is treated as a strategic asset for discovery, signing, marketing, partnerships, live performance and long-term artist development.
- External subscription pricing, public signup and public-facing SaaS features are removed from the core product plan.

1. The business outcome

1.1 Why the label needs this

A&R teams face more music, creators, trends and audience signals than people can consistently evaluate by hand. Headline statistics can mislead: one viral clip can inflate attention without creating durable fans, a track can outperform while the rest of the artist catalogue stays flat, and a large audience can be poorly matched to the artist's intended markets or business opportunities.

Breakout Radar connects the evidence that is usually scattered across data vendors, platform analytics, public content, music releases, live observations and internal scout judgement. It creates a dated, explainable record of **what the team knew at the time**, what it decided, and what happened afterward.

1.2 Business outcomes to optimize

The tool is successful when it improves the label's ability to:

- identify exceptional artists earlier than ordinary market signals would;
- reduce time spent on artists whose apparent traction is fragile or artificially inflated;
- recognize artists with both creative strength and authentic audience development;
- understand which audience segments are becoming fans, not just which counts are increasing;
- choose better release, content, collaboration, territory and live-performance strategies;
- identify viable brand partnerships and monetization paths that fit an artist's audience;
- build lasting artist brands, not just short-lived viral singles;
- learn from signings, passes, development decisions and subsequent outcomes.

The commercial outcome is successful talent discovery and artist development - not software subscription revenue.

1.3 Non-goals

Breakout Radar will not:

- replace A&R judgement or promise that an artist will become a star;
- treat follower count, streaming volume or virality as artistic quality;
- claim that public APIs reveal the identity of individual listeners when they only expose aggregates;
- label an artist a fraud based on a single anomaly;
- present correlation as proof that a marketing tactic caused growth;
- build every advanced audio, video, graph and forecasting capability before the MVP proves value;
- assume that all social or streaming data is available, exportable, retainable or usable for model training.

2. The operating workflow

The product should follow the actual workflow of an in-house label team.

1. **Scan:** ingest approved artist, track, UGC, audience, chart, playlist, release, live and internal data.
2. **Resolve:** map aliases, tracks, sounds, social accounts and releases to a canonical artist identity.
3. **Detect:** find unusual movement in audience, UGC, streaming, catalogue or geography.
4. **Validate:** assess authenticity risk, audience conversion and available evidence quality.
5. **Review the art:** evaluate artistic potential using a structured expert rubric and evidence from multiple works.
6. **Rank:** prioritize eligible artists by a transparent, multi-dimensional A&R score and confidence band.
7. **Decide:** Watch, Monitor, Investigate, Contact, Live Review, Develop, Signing Candidate or Pass.
8. **Develop:** for roster artists, understand audience segments and recommend content, market, partnership and career-development actions.
9. **Record outcomes:** save scout decisions, actions, results and reasons.
10. **Backtest and learn:** compare historic predictions to later outcomes and periodically adjust the models.

The system should maintain separate views for **prospects**, **artists in active A&R evaluation**, and **signed roster artists**. Discovery scoring and roster-development recommendations use related data but are not identical jobs.

3. Product modules

3.1 A&R Radar - the daily priority queue

The homepage answers: **what changed, why does it matter, and what should the team do?**

Each candidate card should show:

- artist name, canonical identity, country, city and relevant genres;
- current market visibility and the date/source of the measurement;
- momentum versus an appropriate peer baseline;
- UGC velocity, creator diversity and content-type signals;
- audience conversion/depth and audience-composition shifts;
- artistic-potential status and whether a human creative review is complete;
- catalogue repeatability and number of tracks contributing to demand;
- geographic movement and likely next markets;
- growth-integrity status and reasons;
- data coverage/confidence;
- recommendation, assigned scout and next action.

The Radar must separate at least four feeds:

1. **Organic Breakout Candidates** - strong evidence, no unresolved high-risk integrity flag, sufficient artistic review.
2. **Promising - Creative Review Needed** - growth is interesting, but artistic evaluation is pending.

3. **Growth Integrity Review** - material suspicion of artificial inflation, identity mismatch or data anomaly.
4. **Roster Audience Opportunities** - current signed artists whose audiences or UGC patterns suggest a development or monetization opportunity.

A questionable signal should not vanish silently. High-risk records are filtered out of the clean organic ranking but remain visible in the integrity-review queue with evidence and an audit trail.

3.2 Artist Intelligence Brief

Every artist profile should be readable in approximately 15 seconds, with deeper evidence available below the fold.

Header - Artist identity, country/city, genre context, career stage and account-verification status. - A&R Priority, Evidence Confidence, Organic Growth Confidence and Artistic Potential status shown as distinct measures.

Core panels - Momentum and expected-performance comparison. - UGC Pulse and top fan-created content. - Audience Morphing: current composition versus previous snapshot. - Virality vs Fandom funnel. - Artistic Potential Review and reviewer disagreement. - Catalogue Health and multi-song conversion. - Geography and audience-affinity map. - Comparables: sonic, audience, career-stage and trajectory. - Risk and missing-data panel. - Internal notes, decisions, tasks, meetings and contact history.

Required explanation: every score must expose its strongest positive signals, key risks, missing evidence, last-updated time, source provenance and recommended next action.

4. UGC Intelligence - a first-class module

4.1 Definition

UGC means **user-generated content made by fans, creators or other third parties that features, uses, references, remixes, performs or responds to an artist or their music**. It is not the same as content posted by the artist's official account.

This distinction matters. Music-intelligence vendors already separate an artist's own TikTok posts from fan-created videos using the artist's sound; their products describe sound-use counts, creator activity, content categories and creator demographics as useful signals for understanding organic spread [R1, R2]. Breakout Radar should capture the same class of signal where licensed and permitted, then connect it to its own audience, artistic and A&R workflow.

4.2 What to discover

For every monitored artist and track, UGC discovery should search and reconcile permitted identifiers and query variants:

- canonical artist name, aliases and common misspellings;
- track title, alternate title, featured artists and released versions;
- ISRC or other track identifiers when available;
- official platform sound/audio IDs and approved sound mappings;
- creator handles and official artist accounts;
- campaign hashtags, challenge names and trend phrases;
- lyrics or short phrases only where lawful, permitted and useful for discovery;

- producer/collaborator names where there is a clear relationship.

A match must have a **UGC attribution confidence**: confirmed official sound, probable track match, artist mention only, or needs human validation. Do not assume that every video using the same or a similar audio snippet belongs to the artist.

4.3 Separate content classes

Classify UGC into actionable categories, for example:

- dance/challenge;
- lip-sync;
- cover or reinterpretation;
- live-performance clip;
- fan edit/tribute;
- reaction/review;
- tutorial/how-to;
- meme or comedic use;
- lifestyle, fashion or transition video;
- sports, gaming or other contextual use;
- creator/artist collaboration;
- brand/commercial use;
- other / unclassified.

Also distinguish original creator content from reposts, compilations and duplicated uploads where the data source allows it. Categories should be probabilistic and include an “unclassified” option; they are not facts when the available evidence is weak.

4.4 UGC metrics to store

For each source, sound, track and artist, create dated snapshots of the fields allowed by the source contract. Useful fields include:

- total observed UGC posts/videos using the sound;
- new unique posts in the period and cumulative count;
- unique creators and first-time creators;
- creator repeat rate and creator retention;
- creator-size tier distribution, where available;
- video/post views, likes, comments and shares, where visible and permitted;
- median engagement, not only sums dominated by the biggest posts;
- UGC growth velocity and acceleration;
- creator and post concentration (e.g. how much activity comes from the top 1/5/10 creators/posts);
- content-type distribution and its change over time;
- language and geography where the source provides reliable aggregates;
- official-sound use versus probable unofficial or re-uploaded audio;
- cross-platform appearance and migration;
- post persistence/decay after a spike;
- UGC per 1,000 audience units only where the denominator is consistent and licensed;

- downstream changes in artist follows, searches, streaming, catalogue activity, ticket interest or first-party sign-ups where observable.

Store the **source, observation time, publication time, collection time and metric definition**. A public count retrieved today is not a historic count from the day the video was published. Data gaps must be explicit.

4.5 The UGC quality lens

One hundred thousand posts driven by a small number of large accounts may represent a different kind of demand from 10,000 posts made by many independent creators across cities and communities. Therefore Breakout Radar must look beyond total UGC count.

The UGC assessment should answer:

1. Is the number of new posts accelerating?
2. Are more distinct creators joining, or are the same accounts carrying the trend?
3. Is creator concentration decreasing or increasing?
4. Is the song being used in multiple contexts or one narrow trend?
5. Is UGC spreading across markets and languages?
6. Are new creators returning to use the sound again?
7. Is the trend causing people to discover the artist, follow them or consume more songs?
8. Is activity persisting after the first viral peak?
9. Is the content apparently fan-led, campaign-led, paid/partner-led, or uncertain?

UGC Virality is not automatically fandom. Strong UGC becomes a stronger A&R signal when it is accompanied by broader creator participation, repeat consumption, profile/follow conversion, catalogue activity or durable audience expansion.

4.6 UGC alerts and actions

Examples:

- **Sound acceleration:** new UGC posts using Track A are 3.2x above the artist's usual 14-day pace.
- **Creator diversification:** activity has moved from three large creators to a broad set of smaller creators.
- **New use case:** the track is moving from dance videos into fashion/lifestyle content.
- **Geographic spread:** UGC creators in Kampala and Dar es Salaam are growing faster than the historical baseline.
- **UGC without conversion:** video creation is increasing but artist follows and other catalogue consumption remain flat.
- **Durability risk:** the entire signal depends on one trend creator or a short-lived challenge.

The recommended action can be to contact creators, brief the artist's team, test a content format, monitor a market or investigate whether a paid campaign explains the spike. The tool must label action recommendations as hypotheses, not as proven causal effects.

4.7 UGC data feasibility

Do not assume one public API exposes every video, creator, sound and audience field. The implementation should use licensed music-intelligence feeds, authorized platform access, official artist/label account connections and human curation where legally appropriate. TikTok's Research API requires qualification and

approval and is explicitly not generally available to commercial users; its commercial-content access is a different, limited product [R6, R7]. This makes provider due diligence a Phase 0 requirement, not an engineering afterthought.

5. Audience Intelligence - the central product capability

5.1 Audience is not one number

The audience is the link between talent and long-term value. The system must distinguish **audience size**, **audience composition**, **audience behaviour**, **audience relationship depth**, and **audience commercial fit**.

Audience coverage varies by platform and provider. Some sources provide age/gender bands, city/country, language or affinity data; other sources provide only broad aggregates or nothing useful for that field. Existing music-intelligence products document this variation and keep source/platform context visible [R3, R4]. Breakout Radar must do the same.

5.2 Audience segment model

Where aggregate data is permitted, build a consistent, versioned profile across these dimensions:

- **Geography:** country, city and region.
- **Age bands:** use source-provided bands; do not infer exact age from unrelated signals.
- **Gender:** only where provided by the source, with unknown/unspecified retained.
- **Language:** audience or creator language where reliably available.
- **Platform:** streaming, short-form video, long-form video, social, live, radio and first-party channels.
- **Discovery source:** UGC, artist posts, collaborators, playlist/chart, radio, live event, paid campaign, search or unknown where measurable.
- **Behaviour:** new, returning, repeat consumption, multi-track consumption, follower conversion, save/share/UGC actions, ticket/merch/newsletter engagement where available.
- **Affinity:** adjacent artists, genres, playlists, creators or interests, but only when supplied lawfully as aggregate information.
- **Artist relationship stage:** new-to-artist, engaged, repeat, core/high-intent, dormant or reactivated. These labels should only be used when the available data supports them; do not pretend cross-platform identities are linked when they are not.

Do not invent audience demographics from artist genre, location or appearance. Unknown is a valid value.

5.3 Audience Morphing - what changed since the last snapshot?

The feature requested by management becomes a core module, not a chart buried in a report.

At every scheduled snapshot, answer:

- Who is listening or engaging now, at an aggregate cohort level?
- Which audience segments gained or lost share?
- Which segments contributed most to total audience growth?

- How different is the audience mix from the previous snapshot and the recent baseline?
- Is the artist reaching more of the same people, a new audience, or a different mix entirely?
- Are the changes persistent or one-off?
- Did the change coincide with a release, collaboration, UGC trend, playlist placement, campaign, radio play or live event?
- What should the label change in content, market focus or development strategy?

Minimum output for each snapshot comparison

1. **Current composition:** segment shares and estimates with source labels.
2. **Change:** percentage-point change and relative change for each segment.
3. **Contribution to growth:** which segments account for the net increase, where absolute counts are available.
4. **Distribution shift:** a statistical measure comparing the current segment distribution to the previous one and to a rolling baseline.
5. **Confidence:** data coverage, sample size limitations, source freshness and whether the shift exceeds normal volatility.
6. **Interpretation:** a short written explanation, with evidence and caveats.
7. **Action:** content, collaboration, territory, live, CRM or monetization experiment to consider.

Example

> **Audience shift detected: Kampala's share of the observed audience rose from 8% to 17% over two snapshots (+9 percentage points). Nairobi still contributes the largest absolute audience, but Kampala contributed the fastest growth. The timing overlaps with an increase in fan-created videos using Track B. Confidence: medium because one source has limited city coverage. Recommended action: validate with the artist's team and run a small Kampala content/live-market test.**

The example is illustrative. The tool must never populate such numbers without real observed data.

5.4 How to quantify audience morphing

Implement the following in stages:

- **Delta in share:** current segment share minus prior share, reported in percentage points.
- **Absolute contribution:** change in estimated segment size where the source supports counts. This distinguishes a growing total audience from a changing composition.
- **Jensen-Shannon divergence (JSD):** a bounded comparison between two segment distributions, suitable as a summary of overall mix change when bins are harmonized. It should be interpreted against the artist's historical baseline, not a universal magic threshold.
- **Population Stability Index (PSI):** an optional operational diagnostic, useful for tracking distribution changes but not a proof that the change is meaningful or harmful. Use with other evidence [R10].
- **Segment lift:** each segment's growth compared with the artist's whole-audience growth and a relevant peer cohort.
- **Persistence:** whether a detected shift survives the next snapshot, rather than disappearing immediately.

The output should show both the whole-audience change and the key segment movements. If the source uses changing age bins, changing geographic coverage or a revised methodology, do not calculate a misleading distribution comparison. Flag the break in series instead.

5.5 Audience cohorts: composition is not individual identity

A public or commercial aggregate audience source will generally not let the label identify exactly which individual listener was listening yesterday. Breakout Radar must not imply that it can. It should compare **aggregate cohorts and distributions**, not reconstruct individual behaviour without an authorized first-party data basis.

Where the label has consented first-party data - such as opt-in mailing lists, ticket purchases, fan-club records, merch orders or artist-site accounts - a separate first-party audience layer can measure lifecycle cohorts and repeat behaviour under documented access controls and purpose limitations. Do not merge a person across platforms unless identity linkage is authorized, necessary and supported by consent/contract.

5.6 Cadence and snapshot policy

Management referred to a “bimonthly” social-number update. That term can mean twice per month or once every two months. The application should support both, but the operating owner must confirm which cadence is intended.

Recommended starting configuration:

- UGC and source-level trend snapshots: daily where provider access and refresh rates allow; otherwise use the source’s actual update schedule.
- High-priority spike alerts: event-driven or daily where available.
- Audience Morphing formal comparison: every 14 days (fortnightly) as a starting default, configurable to every two months if that is the intended business cadence.
- A&R review pack: at each formal audience update and before major release/signing decisions.
- Signed-roster audience review: at a cadence aligned to release cycles and campaign planning.

Keep the cadence configurable per source and metric. Never describe a stale weekly vendor feed as real-time.

5.7 From audience knowledge to artist development

Every audience insight should be capable of producing an action. Examples:

- A new language or geography cluster appears: test appropriate content, collaborations or market visits.
- A segment is large but has low repeat engagement: test stronger artist storytelling, catalogue sequencing or a deeper-content format.
- UGC comes mainly from creators in one scene: identify adjacent creator communities, not only celebrity influencers.
- Live interest rises in a city: validate with promoters, waitlists or a small show before making a large commitment.
- An audience segment overlaps with an adjacent artist: explore a collaboration if creatively appropriate.
- Audience affinity supports a brand category: produce a brand-fit brief based on aggregate evidence, audience values and artist identity, not stereotypes.
- First-party fans are becoming inactive: test an opt-in reactivation campaign and compare outcomes.

Every recommended action should record the hypothesis, chosen segment, intended outcome, spend/time commitment, execution date and result. This is how the label learns which approaches work for which kinds of artists.

6. Growth Integrity - organic, paid, suspicious and artificial

6.1 Do not collapse everything into “organic” or “fraud”

The platform needs to distinguish five states:

1. **Likely organic/authentic:** coherent growth supported by independent creators, repeat behaviour, cross-signal conversion and expected context.
2. **Legitimate paid/partner-assisted:** growth appears connected to documented ads, influencer deals, promotion, radio, campaign spend or other legitimate activity. Paid growth is not automatically artificial.
3. **Mixed or uncertain:** multiple explanations are plausible or the data is inadequate.
4. **Elevated artificial-growth risk:** several independent signals disagree with a credible organic trajectory and require review.
5. **Confirmed invalid/manipulated:** a platform, distributor or approved provider has supplied a confirmed flag, or a documented internal investigation reaches the label’s defined standard.

Spotify warns that services guaranteeing streams or playlist placements are illegitimate and describes patterns including unexplained spikes, sudden geographic changes and a spike followed by a rapid drop; public figures may be adjusted while private analytics can temporarily differ [R5]. This supports a multi-signal review system, not a conclusion from one metric.

6.2 Organic Growth Confidence

Calculate a transparent **Organic Growth Confidence** indicator from multiple signal families. Initial features should include:

- growth relative to the artist’s own historical baseline and a career-stage/genre/territory peer baseline;
- consistency across platforms and metrics that are legally comparable;
- diversity of creators, posts, playlists and sources;
- audience conversion: do streams/views correlate with follows, repeat consumption, saves, shares, catalogue consumption or UGC?
- geographic plausibility relative to known history and campaign activity;
- persistence after the spike;
- UGC creator diversity and concentration;
- relationship between release/campaign events and observed growth;
- known paid spend, influencer work, radio pushes or other legitimate activity;
- platform/distributor warnings or provider-confirmed alerts;
- data reliability, freshness and identity confidence.

Do not use any single ratio as a universal fraud detector. Thresholds must be calibrated per platform, country, genre and artist stage. A large divergence can be caused by an event, a provider refresh, a platform definition change or a new audience market.

6.3 Suspicion signals to investigate

Potential signals include:

- an extreme, unexplained jump relative to the artist’s history and peer cohort;

- a sharp spike followed by an immediate drop;
- a new geographic source inconsistent with known audience and campaign history;
- high consumption with little corresponding downstream artist engagement, where the metrics are comparable;
- sudden follower or engagement increases without plausible creator diversity;
- UGC or playlist activity concentrated in a small set of accounts or placements;
- repetitive, synchronized or otherwise abnormal activity patterns where permitted data supports analysis;
- different platforms moving in contradictory ways without an obvious explanation;
- known third-party stream-guarantee promotions or warnings;
- identity mismatch, suspicious profile duplication or song misattribution.

Counter-signals must be recorded too: marketing spend, official partnerships, music-video releases, media coverage, tour dates, viral creator posts, playlist campaigns and provider data revisions. The goal is to explain the evidence, not punish legitimate growth.

6.4 Operating rule for filtering

- **Confirmed invalid/manipulated:** exclude from clean organic-performance calculations for the affected metrics where the source allows; retain the record and provenance for audit.
- **Elevated risk with multiple independent signals:** remove from the “clean organic breakout” shortlist pending human review; show in the integrity queue.
- **Mixed/uncertain:** down-rank or restrict the recommendation to MONITOR; request additional evidence.
- **Legitimate paid/partner-assisted:** keep eligible but label the growth source and evaluate whether audience conversion persists without that support.
- **Likely organic/authentic:** eligible for normal ranking, subject to artistic review and data confidence.

No artist should be permanently blacklisted by an unreviewed model flag. Every flag needs an explanation, evidence links, reviewer outcome and review date. The model should optimize for catching meaningful risk without turning incomplete data or smaller-market coverage into unfair penalties.

7. Artistic Integrity and Potential - included in Phase 1

7.1 Why this is a hard requirement

Growth alone does not establish that an artist has the underlying craft, identity or repeatable creative ability for a lasting career. A song may perform well because of a meme, a dance or a one-off placement. The label needs to establish that there is an artist worth investing in, not just a signal worth watching.

Artistic potential must therefore be part of the first release. It should be evaluated independently enough from fame metrics to reduce popularity bias, then combined with audience evidence for an overall A&R decision.

7.2 Separate three questions

A. Artistic Potential

Does the work show craft, distinctiveness, expression and signs of repeatability?

B. Artist Identity

Is there a coherent and distinctive musical/public identity that can develop into a recognizable brand?

C. Rights and Authenticity Due Diligence

Are artist identity, credits, authorship/provenance, samples, collaborations and relevant rights claims sufficiently understood to proceed? This is a due-diligence workflow, not a claim that software can prove originality from audio alone.

These dimensions must not be conflated. A rights question does not necessarily mean the music lacks merit; a high engagement score does not prove the music is original or artistically strong.

7.3 Artistic Review Rubric

Use a structured 1-5 panel rubric with examples for each score and mandatory notes. The proposed dimensions are:

1. **Distinctiveness / artistic identity** - does the artist have a recognizable point of view or sound?
2. **Songwriting and composition craft** - melody, structure, lyrical/storytelling choices, arrangement and intentionality, judged in relevant genre/language context.
3. **Emotional impact and communication** - does the performance or recording communicate a clear feeling, perspective or experience?
4. **Performance / delivery** - vocal, instrumental, production or performance strengths appropriate to the artist's chosen lane.
5. **Repeatability across catalogue** - evidence that the strongest qualities recur across more than one song, rather than relying on one outlier.
6. **Evolution and learning potential** - signs of range, development, collaborative ability or a clear path for improvement.
7. **Cultural and audience resonance** - evidence that the work connects with its intended communities, assessed with local/contextual knowledge rather than a single global template.
8. **Artist identity coherence** - whether songs, visual identity, public narrative and live presence are aligned enough to support a credible brand; early-stage gaps can be development opportunities.
9. **Live or performance potential** - where evidence exists; otherwise mark unknown and schedule a live review rather than guessing.
10. **Professional readiness / execution** - reliability, release cadence, collaboration, communication and practical development needs, kept separate from inherent creative worth.

Not every dimension should be equally weighted for every role or genre. The rubric should allow a deliberate specialist to score differently from a songwriter or live-performance expert while preserving a common language.

7.4 Human panel design

For serious candidates:

- obtain at least two independent assessments; use a third assessor for high-stakes or highly divergent cases;
- where practical, conduct an initial blind listening before exposing reviewers to streaming/social numbers;
- review multiple works (ideally three tracks or a track plus other evidence) and a live/performance sample where available and permitted;
- record time-coded or track-specific evidence, not just a score;
- preserve disagreements rather than averaging them away silently;

- calibrate reviewers on sample artists and review inter-rater agreement regularly;
- include reviewers with relevant genre, language and regional knowledge;
- provide a “not enough evidence” option and separate confidence from score.

Research on musical quality evaluation illustrates the value of independent expert ratings and measuring agreement rather than assuming a single evaluator is ground truth [R11]. AI may assist with organizing evidence or extracting permitted measurable features, but it must not be the sole judge of artistic merit.

7.5 AI’s role in creative assessment

Where rights and technical conditions allow, the system may extract measurable descriptors such as tempo, duration, structural cues, instrumentation, vocal/pitch characteristics, genre probabilities, repetition, acoustic similarity and production descriptors. These are **indicators**, not a definitive definition of good music.

AI may also summarize human notes and organize the evaluation. It should not autonomously decide that a song is “bad,” penalize a language/dialect, equate expensive production with talent, or treat genre conventions as a universal quality scale. For multilingual or region-specific music, use reviewers who understand the context and disclose when language/audio models have weaker coverage.

Do not upload audio, lyrics or platform-derived data to third-party AI systems unless the label has permission and the applicable data terms allow that use. Spotify’s developer policy, for example, expressly restricts creating derived listenership metrics and using Spotify content/platform data to train AI/ML [R9]. Licensed sources and direct artist/label-submitted files should be handled under their own terms.

7.6 How creative merit affects ranking

Use a **two-stage decision design**:

Stage 1 - eligibility and evidence checks - resolve artist identity; - assess growth integrity; - identify unresolved rights/provenance issues; - check whether minimum creative-review evidence exists.

Stage 2 - multi-dimensional A&R priority - combine artistic potential, audience quality, momentum, UGC strength, catalogue repeatability, market expansion and commercial/live evidence.

During MVP, a candidate with incomplete artistic evidence should be labeled **Creative Review Needed**, not silently scored as weak. A serious integrity concern creates a hold for due diligence. Do not make a universal hard cutoff until the panel has calibrated its rubric and the retrospective evaluation shows how the score relates to outcomes.

8. The candidate scorecard

8.1 Keep separate scores visible

Every artist should have independent values for:

- **Artistic Potential** - panel score + supporting indicators.
- **Audience Depth / Conversion** - how much attention has become durable artist-level behaviour.
- **Momentum vs Expected** - observed growth relative to a fair peer baseline.
- **UGC Strength / Diversity** - fan-created spread, creator diversity, persistence and conversion.
- **Catalogue Repeatability** - evidence beyond one song.
- **Market Expansion** - new territory/city traction and durability.
- **Commercial / Live Readiness** - observed demand and available commercial evidence.

- **Growth Integrity** - authentic, paid/partner-assisted, uncertain, elevated risk or confirmed invalid.
- **Data Confidence** - source coverage, freshness, identity confidence and history depth.
- **A&R Priority** - operational priority for human review.

Do not make a missing value look like zero. Show “unknown” and reduce confidence when evidence is missing.

8.2 Initial transparent weighting for the pilot

The following is a **starting hypothesis**, not a scientifically validated final formula. Review with the A&R panel and adjust only after backtesting.

Dimension	Pilot weight
Artistic Potential	25%
Audience Depth / Conversion	20%
Momentum vs Expected	15%
UGC Strength / Diversity	15%
Catalogue Repeatability	10%
Market Expansion	10%
Commercial / Live Readiness	5%

Growth Integrity is a separate eligibility/risk control; Data Confidence is shown separately and limits how strongly the product may recommend action. This avoids allowing strong vanity metrics to wash out a serious authenticity issue or using missing data as a hidden penalty.

8.3 A&R recommendation states

- **WATCH:** interesting signal; not enough corroboration yet.
- **MONITOR:** promising movement; gather more time-series and audience evidence.
- **INVESTIGATE:** balanced signals justify deeper A&R review.
- **CONTACT:** enough evidence to reach out, subject to internal strategy.
- **LIVE REVIEW:** performance uncertainty is a key open question.
- **DEVELOP:** the creative potential exists, but the artist needs specific development work.
- **SIGNING CANDIDATE:** strong multi-dimensional evidence and acceptable due-diligence state; requires human decision.
- **HOLD / INTEGRITY REVIEW:** authenticity, identity, rights or data concerns must be resolved.
- **PASS:** lower priority relative to current opportunity cost, with reason logged.

Every decision needs a human owner, rationale, next step and review date.

9. Predictive model strategy - build for accuracy, not impressive demos

9.1 Predict a defined outcome

Do not train a vague model for “superstar potential.” Use measurable outcomes and separate product questions. Examples:

Near-term signal outcomes (30-90 days)

- sustained growth above peer baseline;
- continued UGC creator growth after a spike;
- more songs contributing to consumption;
- audience conversion improvement;
- expansion into another city/country cohort.

Mid-term outcomes (6-12 months)

- repeat release performance;
- sustained artist-level consumption rather than single-track dependence;
- material audience expansion that persists;
- validated live demand or promoter interest;
- verified commercial milestones where data is available.

Longer-term outcomes (12-36 months)

- defined audience or consumption milestone;
- national/regional chart entry;
- repeatable live-ticket demand;
- sustained multi-track catalogue growth;
- brand, publishing, sync or other commercial outcomes.

Signing itself is not the sole ground truth: a signing reflects a label’s strategy, roster needs, budgets and human choices as well as artist potential. Capture it as an outcome, but train/evaluate against multiple observed results to avoid learning only the preferences of past decision-makers.

9.2 Build the dataset at the decision moment

The core training unit should be an **artist snapshot at a known date**. For each prediction, only use data that would have been available to the scout on that date. Store a frozen feature version, source timestamps, model version, predicted probability, recommendation and eventual outcome.

Include diverse examples:

- artists who broke out;
- artists who grew steadily but stayed local;
- viral artists whose interest faded;
- artists with slow-burn growth;
- signed and unsigned artists;

- artists whose opportunity was constrained by missing infrastructure;
- artists from different genres, countries, languages, career stages and gender groups where data quality supports fair comparison.

This is essential to reduce survivorship bias and to test whether the model merely recognizes already-famous artists.

9.3 Feature families

Growth and demand: levels, rates, acceleration, persistence and deviations from baseline.

Audience: segment mix, cohort growth, new/returning behaviour where observable, repeat consumption, conversion, geography and affinity.

UGC: new videos/posts using sounds, distinct creators, creator concentration, use-case diversity, geographic/language spread, UGC velocity/decay and downstream artist-level conversion.

Music and catalogue: number of meaningful releases, multi-track momentum, second/third-release evidence, track concentration, lawful audio descriptors and human review scores.

Career and context: career age, release calendar, collaborations, live events, marketing spend/campaign markers, playlists/charts, radio and territory context.

Trust and coverage: identity-match confidence, source freshness, missing data, provider warnings and integrity-review outcomes.

9.4 Candidate modeling progression

1. **Rules and transparent statistical baselines:** percentile growth, robust anomaly detection, cohort benchmarks and a scorecard. This is the first reference model.
2. **Calibrated tabular model:** logistic regression and gradient-boosted trees (e.g. LightGBM/CatBoost) for initial prediction when the dataset is large and consistent enough.
3. **Time-to-event/survival model:** estimate time to a defined milestone when outcomes are naturally time-dependent and some artists have not yet reached it.
4. **Learning-to-rank:** order candidates according to the actual weekly A&R decision problem once enough recorded decisions and outcome data exist.
5. **Embeddings / multimodal models:** later, only where licensing, sample size and proven value justify the complexity.

Do not start with a large deep-learning model on a small and biased dataset. Establish baseline performance, inspect errors, and add complexity only when it measurably improves useful decisions.

9.5 Model validation controls

- Chronological splits: train on the past and test on later dates.
- Rolling-origin backtesting across multiple historical cut-off dates.
- A final untouched holdout period where sufficient history exists.
- A separate test of generalization to artists not seen during training.
- Leakage checks for normalization, imputation, feature selection, artist status, later-released metadata and any features recorded after the prediction date.
- Evaluation by country, genre, language, gender and career stage where sample sizes allow.
- Calibration: do predicted probabilities align with actual outcome frequency?
- Comparison with simple baselines and human A&R shortlists.

- Versioned data, feature definitions, labels, code, model and metrics.

Randomly shuffling time-indexed snapshots can leak future information into training. Forecasting research recommends chronological and rolling-origin evaluation and warns that even preprocessing can cause leakage if it uses future data [R10].

9.6 Do not report accuracy alone

Because a true breakout is rare, ordinary accuracy can look high even for a model that misses the artists we care about. Report:

- **Precision@K:** proportion of the top K recommendations that later meet a defined target;
- **Recall@K:** share of all identified successful cases found in the top K shortlist;
- **Lift over baseline:** how much better the shortlist is than a transparent baseline or random selection from the eligible pool;
- **Lead time:** how early the model surfaced the artist before the event/milestone;
- **False discovery rate:** how many high-priority recommendations did not meet the target;
- **Calibration / Brier score:** whether probability estimates are trustworthy;
- **Coverage and abstention:** how often the tool has enough data to make a prediction;
- **Subgroup performance:** whether error rates differ materially across markets or categories;
- **Decision utility:** whether the shortlist improves scarce scout time and qualified-investigation rate.

The aim is not a flashy model accuracy percentage. It is a shortlist that finds genuinely valuable candidates earlier, without drowning the team in false positives.

10. Backtesting - first build priority

10.1 Management question: how soon?

Target: first backtesting harness within 10 working days; first baseline backtest report by weeks 3-4; stronger model comparisons by weeks 6-8. These are build estimates assuming access to historical data, one product/data owner and one engineer/analyst. If source history does not exist, the harness can still be built, but a reliable historical result cannot be invented.

10.2 Backtesting delivery plan

Time	Deliverable	What it proves
Days 1-3	Data inventory, metric definitions, outcome/target agreement	What historic data exists and what can be validly tested
Days 4-7	Dated artist snapshots, source lineage and target-label draft	The team can reconstruct what was known at each cut-off date
Days 8-10	Backtesting harness + rule-based baseline + first evaluation notebook/ report	End-to-end historical test can run without future data leakage
Weeks 3-4	First rolling-origin baseline report, top-K shortlist review and error analysis	Whether simple signals are useful enough to continue

Time	Deliverable	What it proves
Weeks 5-6	Cohort-specific tests, missing-data stress tests, artistic-review integration	Which segments and features are reliable; where the model fails
Weeks 7-8	Candidate ML comparison, calibration and shadow-mode ranking	Whether ML improves over transparent rules/human triage
Weeks 9-12	Pilot sign-off, monitoring plan and versioned model card, if evidence supports it	Whether the system is fit for decision-support deployment

The weeks are a plan, not a guarantee: historical coverage, provider contract and label availability determine the schedule.

10.3 Historical data limitation

A backtest needs the metrics that existed at the historical prediction date, not just today's counts or an artist's current profile. Some providers have deep time series, some have partial history, and some metrics cannot be reconstructed retrospectively. For every feature, record earliest available date, snapshot completeness and whether historic values are observed or estimated. Never pass current audience demographics off as the demographics from 18 months ago.

Where historical data is missing:

1. backfill only what the source legally provides with timestamps;
2. mark its coverage and confidence;
3. start scheduled snapshots immediately;
4. run a baseline backtest on the valid subset;
5. continue prospective testing while longer-term outcomes mature.

10.4 The first backtest should be simple

Start with a transparent rule-based candidate ranking and a small set of near-term outcomes. Test whether the top-ranked historical candidates showed persistent growth, multi-track expansion, creator diversity and downstream audience conversion over the next 30-90 days. This does not prove the system can predict a 24-month signing success, but it gives the team an early empirical checkpoint before investing in complex ML.

Use 12-24-36 month career milestones as longer-term targets when the data supports them. They will mature later, so the early system must label short-horizon signal validation separately from long-horizon breakout prediction.

10.5 Human shadow evaluation

Before the recommendations influence signing decisions, run the tool in **shadow mode**:

- save the platform's ranked list before scouts make their selection;
- save the A&R team's existing shortlist independently;
- have reviewers assess both, ideally without knowing which source produced a given item;
- track discoveries, misses, time-to-review and reason for disagreement;
- only then decide whether the system improves the workflow.

This creates the evidence needed to judge the tool without letting an unvalidated score dictate label decisions.

11. Internal A&R pipeline and artist development

11.1 Candidate pipeline

`Discovered -> Watch -> Investigate -> Contact -> Creative / Live Review -> Development Plan -> Signing Decision -> Signed / Pass`

A signing candidate must have an evidence pack containing:

- artistic panel assessments and raw reviewer notes;
- audience composition and trend history;
- UGC trajectory and creator ecosystem;
- organic-growth assessment and any unresolved flags;
- catalogue repeatability;
- geography and likely market opportunities;
- comparable trajectories and uncertainty;
- commercial scenarios and development needs;
- due-diligence checklist;
- proposed next action and accountable owner.

11.2 Signed-roster development workspace

For signed artists, switch the central question from “Should we sign?” to **“How do we serve this audience, strengthen the artist’s brand and grow a durable business?”**

The workspace should support:

- audience segment briefs and changes over time;
- fan-content and creator strategy;
- release and catalogue conversion monitoring;
- territory expansion plans;
- collaboration opportunities;
- live and promoter opportunities;
- opt-in first-party fan development;
- merch, ticketing, publishing, sync and partnership hypotheses;
- brand-fit analysis based on aggregate audience data and the artist’s own identity;
- experiment tracking and post-campaign readouts.

Do not imply monetization is guaranteed. Present options, assumptions, confidence and measurable tests.

11.3 Internal collaboration

Each artist record should support:

- scout owner and second reviewer;
- notes and attachments;

- meetings and contact log;
- tasks/deadlines;
- release and campaign calendar;
- internal status changes;
- confidential score and decision history;
- role-based visibility;
- approval steps for sensitive integrity findings and signing recommendations.

12. MVP scope - revised to include management's requirements

The first release must prove the core loop: **find a promising artist, verify authentic growth, assess artistic potential, understand the audience and decide what the team should do.**

Phase 0 - data and decision foundations (Weeks 0-2)

- Name an owner for product, data access, A&R rubric and decision targets.
- Inventory every potential source, coverage, cost, refresh, history depth, data rights and retention rules.
- Confirm the meaning of "bimonthly" for formal social updates.
- Define artist identity, track/sound matching and source lineage.
- Define the first artistic-review rubric and panel calibration sample.
- Agree a small set of 30/90-day targets plus 12-month outcomes.
- Start storing timestamped snapshots from day one.
- Build a gold-standard manually reviewed artist sample for comparison.

Phase 1 - Internal A&R MVP (target 8-12 weeks)

Must ship

1. Canonical artist and track identity resolution with match confidence.
2. Multi-source time-series metrics and source freshness/provenance.
3. UGC Intelligence: sound use, distinct creators, content categories, creator concentration, UGC velocity, trends and conversion signals where data access permits.
4. Audience Morphing: previous/current segment comparisons, growth contribution, change measures, confidence and written interpretation.
5. Audience quality and fan-conversion funnel.
6. Growth Integrity: organic/paid/mixed/suspicious/confirmed status, risk evidence, human-review queue and clean-organic filter.
7. **Artistic Potential Review in Phase 1:** structured rubric, multiple reviewers, catalogue samples, score disagreement and rights/provenance status.
8. Catalogue repeatability: measure whether demand reaches more than one track.
9. Expected-performance and momentum baselines.
10. Transparent pilot A&R scorecard with separate artistic, audience, UGC, integrity and confidence dimensions.

11. Backtesting harness available by day 10, with first baseline report aimed at weeks 3-4 if source history supports it.
12. Radar, Artist Intelligence Brief, alerts, search, watchlists and assigned next actions.
13. Basic internal A&R pipeline, notes and feedback capture.
14. Role-based access, audit log and data-rights metadata.
15. A&R Copilot for evidence-grounded summaries and natural-language queries, limited to data actually present in the internal system.

Explicitly not required for first release

- fully automated audio/video creative judgement;
- a graph neural network or giant knowledge graph;
- automatic contract valuation;
- fully automated market-entry budget decisions;
- a general-purpose agent with write access to sensitive systems;
- all platforms/countries on day one;
- public user accounts, public pricing or multi-tenant SaaS architecture.

Phase 2 - after MVP evidence

- richer licensed audio/music analysis;
- full audience-affinity graph where data rights permit;
- deeper live-demand data;
- advanced artist trajectory/survival models;
- structured campaign experiments and uplift analysis;
- better creator/UGC network analysis;
- brand partnership workflows;
- more extensive first-party CRM integrations;
- multilingual content analysis improvements;
- improved predictive calibration as outcomes accrue.

Phase 3 - strategic intelligence

- multimodal artist and cultural-trend analysis;
- network and scene discovery across artists, producers, creators, venues and cities;
- advanced catalog and repertoire development recommendations;
- portfolio-level planning across the label roster;
- commercial scenario modelling using verified internal and licensed data.

The phases are based on evidence and value, not technology novelty. Creative assessment, UGC, audience change, growth integrity and early backtesting are not postponed to Phase 2.

13. Data and platform constraints

13.1 Provider abstraction is mandatory

Build an internal data-service layer with interchangeable provider adapters. Do not hard-wire the application to one vendor or assume an API field will remain available forever.

```
Approved data sources
  - licensed music-intelligence provider(s)
  - authorized artist / label account connections
  - approved public/platform APIs
  - approved UGC/content provider
  - artist/manager submissions and internal A&R observations
  - internal first-party fan, ticketing, merch and campaign data
              |
              v
Provider adapters + rights/coverage registry
              |
              v
Identity resolution -> timestamped snapshots -> feature store
              |
              v
Audience / UGC / integrity / creative review / prediction
              |
              v
Private A&R workspace + alerting + audit log
```

13.2 Data licensing decisions

For every field or feed, document:

- approved business purpose;
- contract/terms and territory coverage;
- whether it can be stored historically;
- permitted refresh/caching period;
- whether it can be used to calculate derived metrics;
- whether ML training is permitted;
- whether internal employees may view/export it;
- whether derived scores may be retained after source data expiry;
- takedown/deletion and refresh obligations;
- API quotas, refresh latency and reliability.

Spotify’s developer policy restricts analysis for derived listenership metrics and prohibits using Spotify content/platform data to train AI/ML [R9]. Internal use does not automatically remove those restrictions. The label must use contractually suitable licensed feeds for analytics and model training rather than assuming that a public API is sufficient.

13.3 Source reality

- **Music-intelligence vendors:** potentially the best route for cross-platform history, sound use and audience aggregates; validate African coverage, API rights, backfill and cost in a real vendor proof-of-concept.
- **YouTube Data API:** can discover matching videos/channels and obtain documented metadata; it has quotas and is not a universal historical analytics warehouse [R8].

- **Instagram APIs:** professional-account access and permissions have limits; access to consumer accounts is not universal [R12].
- **TikTok:** Research API access requires eligibility and approval and is not generally available to commercial users; commercial-content APIs have their own product scope and geographic limitations [R6, R7]. Contracted third-party data may be required for broader commercial analysis.
- **MusicBrainz:** useful open metadata for artist, recording, release, work and identifier resolution; it is not a source of audience analytics [R13].
- **First-party label data:** valuable for verified artist accounts, campaign events, budgets, opt-in audiences, ticketing/merch and actual business outcomes; apply role permissions and data minimization.

The first technical milestone is a source audit and vendor test, not building every scraper.

14. Privacy, governance and access control

The tool is internal, but internal does not mean unrestricted.

- Use aggregate audience segments wherever individual identity is unnecessary.
- Use opt-in first-party records only for stated purposes and under appropriate consent/access rules.
- Do not attempt to re-identify anonymous listeners or infer sensitive personal traits.
- Limit access to individual contact/CRM data to staff who need it.
- Keep source provenance, collection time, permitted purpose and deletion rules alongside the data.
- Define retention periods for raw posts, snapshots, internal notes and first-party records.
- Restrict exports of sensitive integrity-review material.
- Log who viewed or changed scores, risk flags, panel assessments and signing recommendations.
- Give staff a way to record corrections to artist identity, campaign context or false-positive flags.
- Have legal/data-protection counsel review applicable Kenyan and other market privacy, copyright, database and platform requirements before production use.

The product should help the label make better decisions while respecting audiences, artists and creators.

15. Recommended interface and private navigation

The product should feel like a premium internal intelligence workspace - not a public marketing website.

Main navigation

- **Radar** - daily candidate priorities and alerts.
- **Artists** - prospect database and canonical artist profiles.
- **UGC Intelligence** - sounds, posts, creators, content contexts and growth.
- **Audience Lab** - morphing, segments, affinity and growth contributions.
- **Creative Review** - panel rubric, evidence, disagreements and review status.
- **Integrity Queue** - suspicious growth, identity/rights review and dispositions.
- **Markets** - city/country trends and expansion opportunities.

- **Release Monitor** - release comparisons and catalogue conversion.
- **Backtest Lab** - historical cut-off, rankings, metrics, error analysis and model versions.
- **A&R Pipeline** - contact, audition, development, signing and roster workflows.
- **Roster Studio** - audience development, content strategy and commercial opportunities for signed artists.
- **Research Copilot** - read-only questions and summaries grounded in internal evidence.
- **Admin & Data Health** - access, provider status, source coverage and audit logs.

Visual direction

Use a restrained premium dark/light system, high-contrast typography, artist photography, readable signal cards, precise charts, audience-shift views and UGC thumbnails where licensed. Avoid flashy gradients that hide evidence. Use explicit labels for observed data, derived metrics, model inference and human opinion.

Every important screen must make three things clear:

1. **What changed?**
2. **Why does it matter?**
3. **What should we do next?**

Recommended Radar layout

Top row: candidates requiring action, new UGC acceleration, audience shifts, integrity reviews, upcoming A&R tasks.

Main panel: ranked candidates with A&R priority, artistic-review status, UGC pulse, audience conversion, organic-growth confidence and data confidence.

Side panel: market movers, emerging city/genre signals, key roster audience opportunities and missing-data warnings.

Avoid implying that example/demo numbers are real observations. Demo mode should be clearly marked.

16. Data model - minimum viable entities

The first schema should model the label's decisions as well as the market signals. Keep identity records, observations, model scores and human decisions separately versioned.

Domain	Core entities
Artist identity and catalogue	`artists`, `artist_aliases`, `artist_accounts`, `artist_identity_matches`, `tracks`, `recordings`, `releases`, `track_artist_relationships`
Source registry and metrics	`platforms`, `providers`, `source_rights_registry`, `metric_definitions`, `metric_snapshots`, `artist_time_series`, `track_time_series`, `source_freshness_events`
UGC intelligence	`ugc_items`, `ugc_item_snapshots`, `ugc_sound_matches`, `ugc_creator_entities`, `ugc_content_categories`, `ugc_campaign_events`, `ugc_attribution_confidence`
Audience intelligence	`audience_snapshots`, `audience_segment_definitions`, `audience_segment_values`, `audience_shift_events`, `audience_affinity_aggregates`, `first_party_opt_ins`
Music-market events	`release_events`, `playlist_events`, `chart_events`, `radio_events`, `live_events`, `campaign_events`
Creative review and integrity	`artistic_reviews`, `artistic_review_dimensions`, `reviewer_assignments`, `reviewer_notes`, `rights_provenance_checks`, `integrity_signals`, `integrity_cases`, `integrity_dispositions`
Scoring and prediction	`feature_snapshots`, `peer_baselines`, `scores`, `recommendations`, `alerts`, `model_versions`, `model_predictions`, `prediction_outcomes`, `backtest_runs`, `backtest_metrics`
A&R workflow	`watchlists`, `scout_notes`, `tasks`, `meetings`, `contacts`, `a_and_r_pipeline`, `roster_development_plans`, `campaign_experiments`, `human_feedback`, `audit_events`

Every observation should preserve at minimum: entity ID, metric/feature name, value, source, source timestamp, collection timestamp, definition/version, territory/coverage and rights metadata. Keep current values and history separate; never overwrite historical records in place.

17. Recommended technical stack

Select the stack to support reliable time-series ingestion, auditability and analysis rather than for trendiness.

- **Frontend:** Next.js + TypeScript.
- **UI:** Tailwind CSS + shadcn/ui or equivalent internal design system.
- **Charts:** ECharts or Recharts; maps only where the data supports geographic precision.
- **Backend:** Python + FastAPI.
- **Database:** PostgreSQL.

- **Time series:** PostgreSQL/TimescaleDB initially; move only when volume requires it.
- **Analytics/ML:** Python, pandas/Polars, scikit-learn, LightGBM/CatBoost when justified.
- **Model tracking:** versioned datasets, features, experiment runs, model cards and reproducible metrics.
- **Vector search:** pgvector if comparables/search use cases prove value.
- **Jobs:** queue/worker framework or Temporal for resilient scheduled ingestion.
- **Object storage:** approved object storage for assets where rights permit retention.
- **Auth:** SSO or enterprise-ready role-based authentication.
- **Observability:** structured logs, error tracking, source freshness, ingestion health and data-quality alarms.

Use a provider adapter pattern so a data-vendor change does not require rebuilding business logic. Keep the first architecture modular but avoid microservices overhead before load or team size requires it.

18. Metrics that prove the tool is useful

Discovery and A&R

- Precision@K for shortlisted candidates.
- Qualified investigation rate.
- Median time from first system signal to human investigation.
- Lead time before defined breakout milestones.
- Rate of strong artists missed by the system, with reviewed reasons.
- Scout time saved per qualified candidate.
- Signing and development outcomes tracked over time, without treating signing as the sole ground truth.

Audience and UGC

- UGC new-post velocity and distinct-creator growth.
- Creator concentration and category diversity.
- Share of UGC signals followed by measurable artist-level conversion.
- Audience composition shifts confirmed in subsequent snapshots.
- Forecasted audience shifts versus observed shifts.
- Number of decisions/experiments informed by Audience Lab and their outcomes.

Integrity

- Rate of flagged candidates confirmed/rejected after review.
- False-positive rate by source/market/genre where sample sizes support analysis.
- Time to resolve a material integrity concern.
- Share of high-risk candidates excluded from clean organic ranking pending review.
- Evidence and reviewer completeness.

Artistic potential

- Panel completion rate.
- Inter-rater agreement and disagreement resolution.

- Relationship between early panel scores and later repeatability/catalogue outcomes.
- Frequency with which AI-derived indicators disagree with expert judgement.
- Representation and error checks across genres, languages and markets.

Model and data quality

- Data coverage, freshness and source outage rate.
- Identity resolution precision and manual-correction rate.
- Model calibration, top-K lift, lead time and subgroup performance.
- Backtest reproducibility and feature leakage checks.
- Proportion of predictions made with adequate confidence versus abstention.

19. Risks and mitigations

Risk	Mitigation
UGC access is incomplete or costly	Audit vendor coverage and rights first; maintain approved manual/first-party alternatives; show missing coverage explicitly.
Public metrics do not reveal individual audience identity	Use aggregate cohort shifts; use first-party identities only with proper consent and authorization.
Viral event mistaken for durable fandom	Require cross-signal conversion, repeat usage, multi-track evidence and persistence.
Legitimate paid campaign mistaken for fraud	Record campaign context; separate paid-assisted from artificial; require multiple signals and human review.
Smaller markets have weaker data coverage	Show coverage/confidence; compare against appropriate peers; do not let absent data masquerade as poor performance.
Artistic taste is reduced to a model score	Include independent human panels, written evidence and uncertainty; AI is an assistant, not the final judge.
Past signings bias the model	Include signed/unsigned successes and failures; define external outcomes; compare model rankings with human shortlists.
Historical backtest is invalid	Preserve dated snapshots; use chronological rolling-origin tests; prohibit future-derived features.
Platform policy changes break features	Provider abstraction, terms registry, data lineage and regular access/contract review.
Confident language creates false certainty	Separate score from confidence, show risks and missing evidence, require human approval for high-stakes actions.

20. First 30 days - practical action plan

Week 1: align and inventory

- Confirm whether “bimonthly” means fortnightly or every two months.
- Name an A&R product owner, data owner, creative-review lead and legal/data contact.
- Inventory data providers, artist accounts, historic snapshots, UGC feeds, audience fields, contract rights and refresh cadence.
- Agree first target outcomes and the definition of an eligible artist.
- Select 50-100 artists for a manually reviewed pilot sample, spanning known breakouts, non-breakouts, viral one-offs and artists from different markets/genres. The number is a working pilot suggestion, not a statistical guarantee.

Week 2: build the minimum data foundation

- Create canonical artist/track IDs and entity match review.
- Ingest first approved time-series and UGC records.
- Save audience and UGC snapshots with metric provenance.
- Implement the first backtest harness and transparent rules-based baseline by day 10 if data access permits.
- Draft and pilot the creative review rubric with independent assessors.

Weeks 3-4: first evidence checkpoint

- Produce the first baseline backtest report where historic coverage exists.
- Launch UGC Pulse and Audience Morphing comparisons on the pilot sample.
- Run the first organic-growth review queue with documented counter-signals.
- Compare top-ranked candidates against the human A&R shortlist.
- Report data gaps, wrong matches, false positive patterns and the next required source investment.

Weeks 5-8: harden

- Calibrate artistic rubric and reviewer agreement.
- Add temporal rolling-origin tests and subgroup checks.
- Compare transparent baselines with one calibrated ML candidate only if the data volume supports it.
- Launch shadow-mode score snapshots and human feedback.
- Test audience/UGC recommendations against actual campaign, release and live actions.

Weeks 9-12: pilot operational use

- Train the A&R team on score meaning, confidence and review queues.
- Deploy role-based workflows and audit logs.
- Approve only the model behavior that has demonstrated value over baseline; keep underperforming modules in monitoring mode.
- Agree the next phase from evidence, not assumption.

21. Final product definition

Breakout Radar is the label's internal A&R intelligence and artist-development system. It combines historically grounded growth signals, fan-created UGC, audience-composition change, organic-growth verification, structured artistic assessment and time-aware prediction to find exceptional talent early and help the label understand, serve and monetize the audiences that form around that talent.

Its operating principle is:

> **The audience tells us where demand is forming. The art tells us whether the artist is worth investing in. Integrity checks tell us whether the signals can be trusted. Backtesting tells us whether our system actually works. Human A&R turns that evidence into decisions.**

Its moat is not a prettier dashboard. It is the compounding dataset linking **artists, music, fan-made content, audience cohorts, creative assessment, growth authenticity, market changes, A&R decisions and real outcomes** - captured with dates, provenance, permissions and reviewable evidence.

The one-sentence build brief

> **Build a private, evidence-first A&R command center that finds artists with exceptional creative potential and authentic audience momentum, tracks how UGC and audience cohorts are evolving, filters suspicious growth out of the organic shortlist, and continuously tests whether its recommendations identify valuable talent earlier.**

> **RESEARCH NOTE | Vendor documentation confirms capabilities and policy constraints, not guaranteed data coverage for this label. Verify territory coverage, retention rights, history depth and ML permissions during procurement.**

Research notes and source register

The product recommendation above separates market/vendor evidence from the label-specific design decisions. Vendor pages are evidence that these signal types exist in commercial music intelligence, not proof that any one provider has the exact African coverage or contractual rights this label needs. Those must be verified in a proof-of-concept and contract review.

[R1] Chartmetric Help Center - How to Monitor TikTok Sounds & Fan Activity with Chartmetric (updated June 25, 2026). Describes a separate sound-use view for fan-created content, creator tiers, content categories, geographic breakdowns and creator demographics. https://support.chartmetric.com/en/articles/98

[R2] Soundcharts - TikTok Analytics. Describes tracking videos using a track, video-count growth, audience demographics/geography and alerts; use as a vendor capability reference to validate in a procurement trial. https://www.soundcharts.com/en/tiktok-analytics

[R3] Chartmetric - Audience Insights. Describes source-dependent demographic, language, geography and affinity dimensions; supports keeping the source and scope visible. https://chartmetric.com/use-cases/audience-insights

[R4] Soundcharts Help Center - The Audience Tab. Describes audience metrics and time-series/fan-base history across available sources. https://help.soundcharts.com/en/articles/3192461-the-audience-tab

[R5] Spotify for Artists - Artificial Streaming. Describes stream manipulation, warning signs and potential platform/distributor actions. It is platform-specific guidance, not a universal fraud classifier. https://artists.spotify.com/en/artificial-streaming

[R6] TikTok for Developers - Research API FAQ (updated September 1, 2026). Explains eligibility and approval requirements and states that creators/advertisers/commercial users are not eligible for Research Tools access under that route. https://developers.tiktok.com/docs/en/research-api-faq

[R7] TikTok for Developers - Commercial Content API: Query Commercial Content. Documents a separate commercial-content API and its specific scope/fields. The product page describes its rollout as limited to EU-country data at the time accessed; do not assume it provides general African UGC coverage. https://developers.tiktok.com/docs/en/commercial-content-api-query-commercial-content https://developers.tiktok.com/products/commercial-content-api

[R8] Google for Developers - YouTube Data API: Search list. Documents search behavior and API quota; search results are discovery metadata, not a complete historical social-intelligence feed. https://developers.google.com/youtube/v3/docs/search/list

[R9] Spotify Developer Policy (effective May 15, 2025, as displayed when accessed October 9, 2026). Restricts analysis to create derived listenership metrics and the use of Spotify content/platform data for AI/ML training. Verify current terms and legal approval before integrating. https://developer.spotify.com/policy

[R10] “Forecast evaluation for data scientists: common pitfalls and best practices,” Data Mining and Knowledge Discovery / PMC (2022). Discusses fixed-origin and rolling-origin evaluation, temporal order and data leakage. Supports chronological backtesting rather than random splits for future-oriented models. https://pmc.ncbi.nlm.nih.gov/articles/PMC9718476/

[R11] Frontiers in Psychology - blind-listening evaluation of AI-generated, human-AI collaborative and human-composed music (2026). Reports expert evaluation and inter-rater agreement for defined music-quality dimensions; supports structured panel design but does not create a universal measure of artistic greatness. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2026.1903971/full

[R12] Meta Instagram API documentation collection on Postman. Describes professional-account requirements and limitations, including lack of access to ordinary consumer accounts through that API route. https://www.postman.com/meta/instagram/documentation/6yqw8pt/instagram-api

[R13] MusicBrainz API documentation. Provides open music metadata resources for artist, recording, release, work and identifier matching; not an audience-analytics data source. https://musicbrainz.org/doc/MusicBrainz_API

[R14] “Dataset shift assessment measures in monitoring predictive models” (ScienceDirect, 2021). Discusses population stability and model-impact measures as complementary diagnostics for distribution shifts; it supports using mix-change statistics as monitoring aids, not as standalone judgements. https://www.sciencedirect.com/science/article/pii/S1877050921018512

Research access date: October 9, 2026. Platform policies, API fields, commercial availability and vendor products can change; re-validate them during implementation and procurement.